Module 4 · Hybrid Retrieval & Grounded Reasoning

RECALL · POST /api/v2/recall

Pi · E5 / 13 October
“What do you remember about where I live, my previous city, and my work?”

user:maya
query: E5

Request + time filters
namespace · bounds · as_of
Optional rules / model planner

query text

Lexical search
PostgreSQL full-text

ranked IDs

PostgreSQL + pgvector
assertions · sources · chunks
namespace + status + time filters

query → vector

nearest IDs

lexical ranks

Pi automatic recall
temporal:false · include_raw:true · max_distance:0.45
top_k:10 · max_tokens:2,048

Semantic search
qwen3-embedding-cpu
pgvector · 1,024-d cosine

semantic ranks

Rank fusion
RRF: Σ 1/(60 + rank)
Deduplicate IDs

Original E1 / E3 evidence

top lexical + semantic IDs
+ query entity matches

Source-text retrieval
Lexical + vector
include_raw / fallback

if temporal enabled
+ time bounds

Temporal candidates
as_of / from / to

Neo4j expansion
Lexical / semantic / entity seeds
MENTIONS + SUPPORTS

IDs + graph paths

Yes · graph ranks

temporal ranks

Eligible graph hit?
PostgreSQL recheck
status · time · distance

Current: R2 Bengaluru · active. **Historical as_of:** R1 Chennai when its interval contains the requested time.
Pi’s automatic temporal:false path can still retrieve history from original source text.

source ranks · E1 / E3

No → discard

Ranked hits + sources

Trace + degraded reasons
Embedding / graph failure or projection lag is reported.
Available retrieval channels continue.

Pi answers the user
Uses relevant recalled evidence
with the current conversation

Relevant memory

Pi context injection
Nonempty + same namespace
Evidence, not instructions

sdp-memory-context

Cited context
[R2] Maya lives in Bengaluru
fact · active · from 10 Oct
Evidence: E3 user quote

IDs · dates · quotes

Evidence packer
top_k + token budget
Up to 3 quotes / hit

Explicit reflect request
uses recall evidence

No evidence → insufficient evidence
Invalid citations → error

Reflection is separate from Pi’s automatic recall.

Grounded answer
“Maya lives in Bengaluru.” [R2]

Yes

Valid citations?
IDs belong to recall hits

answer + citation IDs
+ insufficient_evidence

Optional reflection
mem-extractor:general-v1
POST /api/v2/reflect